

users		
id	uuid	
full_name	varchar(100)	NN
email	varchar(150)	NN
phone	varchar(20)	NN
cnic	varchar(20)	NN
password_hash	text	NN
pin_hash	text	NN
role	user_role	E NN
is_frozen	boolean	NN
is_verified	boolean	NN
avatar_url	text	
created_at	timestamp	NN
updated_at	timestamp	NN
last_login	timestamp	

wallets		
id	uuid	
user_id	uuid	NN
balance	decimal(14,2)	NN
currency	varchar(5)	NN
created_at	timestamp	NN
updated_at	timestamp	NN

transactions		
id	uuid	
sender_id	uuid	NN
receiver_id	uuid	NN
amount	decimal(14,2)	NN
note	text	
status	txn_status	E NN
risk_score	decimal(5,2)	
is_fraud	boolean	NN
fraud_reasons	text[]	
device_ip	varchar(50)	
device_type	varchar(20)	
location	varchar(100)	
hour_of_day	smallint	
sender_txn_count	integer	
receiver_txn_count	integer	
created_at	timestamp	NN
updated_at	timestamp	NN

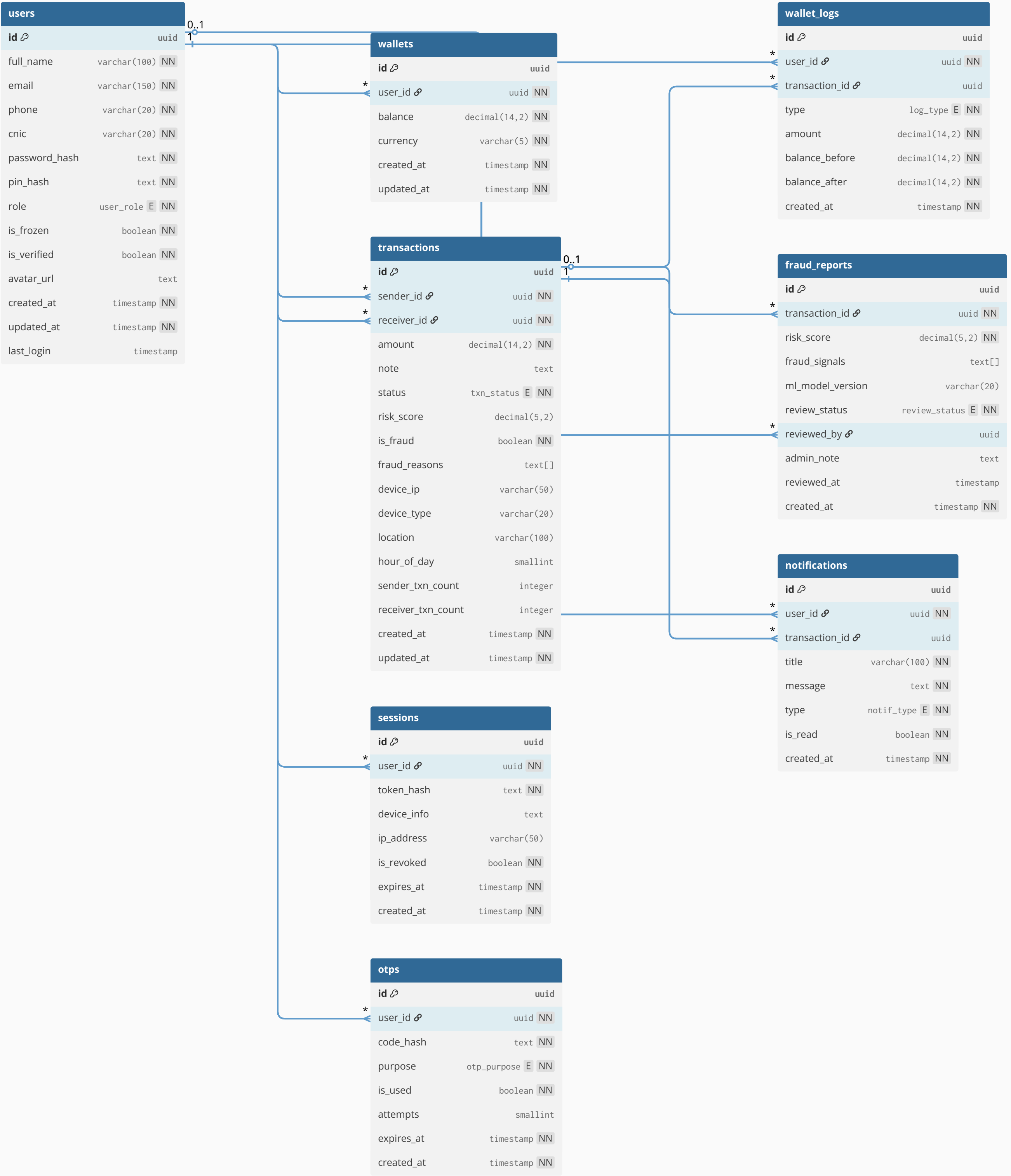
sessions		
id	uuid	
user_id	uuid	NN
token_hash	text	NN
device_info	text	
ip_address	varchar(50)	
is_revoked	boolean	NN
expires_at	timestamp	NN
created_at	timestamp	NN

otps		
id	uuid	
user_id	uuid	NN
code_hash	text	NN
purpose	otp_purpose	E NN
is_used	boolean	NN
attempts	smallint	
expires_at	timestamp	NN
created_at	timestamp	NN

wallet_logs		
id	uuid	
user_id	uuid	NN
transaction_id	uuid	
type	log_type	E NN
amount	decimal(14,2)	NN
balance_before	decimal(14,2)	NN
balance_after	decimal(14,2)	NN
created_at	timestamp	NN

fraud_reports		
id	uuid	
transaction_id	uuid	NN
risk_score	decimal(5,2)	NN
fraud_signals	text[]	
ml_model_version	varchar(20)	
review_status	review_status	E NN
reviewed_by	uuid	
admin_note	text	
reviewed_at	timestamp	
created_at	timestamp	NN

notifications		
id	uuid	
user_id	uuid	NN
transaction_id	uuid	
title	varchar(100)	NN
message	text	NN
type	notif_type	E NN
is_read	boolean	NN
created_at	timestamp	NN



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```
Enum user_role {
  user [note: "Regular wallet user"]
  admin [note: "Can review fraud, freeze accounts"]
}
```

```
Enum txn_status {
  PENDING [note: "ML check in progress"]
  APPROVED [note: "Transferred successfully"]
  BLOCKED [note: "Blocked by fraud detection"]
  FAILED [note: "Insufficient balance or other error"]
  REVERSED [note: "Admin reversed a transaction"]
}
```

```
Enum log_type {
  DEBIT [note: "Money left this wallet"]
  CREDIT [note: "Money entered this wallet"]
  TOPUP [note: "Mock top-up from payment gateway"]
  REFUND [note: "Reversed transaction credit"]
}
```

```
Enum review_status {
  PENDING [note: "Admin has not reviewed yet"]
  CONFIRMED_FRAUD [note: "Admin confirmed it was fraud"]
  FALSE_ALARM [note: "ML was wrong, admin cleared it"]
}
```

```
Enum notif_type {
  SUCCESS [note: "Transaction approved"]
  WARNING [note: "Suspicious activity warning"]
  FRAUD_ALERT [note: "Transaction blocked"]
  INFO [note: "General information"]
}
```

```
Enum otp_purpose {
  PHONE_VERIFY [note: "Verify phone on registration"]
  LOGIN [note: "2FA on login"]
  RESET_PIN [note: "Reset transaction PIN"]
  RESET_PASSWORD [note: "Reset account password"]
}
```